

Software and Hardware Tools

Task 1 :- Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

ANS :-

1. Install: Download and open [Wireshark](#).
2. Capture: Select your Wi-Fi, click the blue shark fin icon to start, browse YouTube/Instagram for 2 minutes, then click the red square to stop.
3. Save: Click File > Save As and save as a .pcap file.

Top 3 Protocols :-

- QUIC / UDP : Used for ultra-fast, smooth video streaming.
- TLS / HTTPS : Encrypts your connection so nobody can snoop on your activity.
- DNS : Converts site names (like youtube.com) into numerical IP addresses.

Task 2 :- Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

ANS :-

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Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\asvin>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42:31::810] with 32 bytes of data:
Reply from 2a04:4e42:31::810: time=34ms
Reply from 2a04:4e42:31::810: time=60ms
Reply from 2a04:4e42:31::810: time=44ms
Reply from 2a04:4e42:31::810: time=45ms

Ping statistics for 2a04:4e42:31::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 34ms, Maximum = 60ms, Average = 45ms

C:\Users\asvin>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [2600:140f:7200:49::17d4:65] with 32 bytes of data:
Reply from 2600:140f:7200:49::17d4:65: time=27ms
Reply from 2600:140f:7200:49::17d4:65: time=57ms
Reply from 2600:140f:7200:49::17d4:65: time=63ms
Reply from 2600:140f:7200:49::17d4:65: time=52ms

Ping statistics for 2600:140f:7200:49::17d4:65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 27ms, Maximum = 63ms, Average = 49ms

C:\Users\asvin>
```

Task 3 :- Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

ANS :-

Application	Process Name	Function	Typical Memory Usage
Google Chrome	chrome.exe (or Google Chrome)	Main Browser Engine	~350 MB
Google Chrome	chrome.exe (Renderer / Tab)	Active Web Page Tab	~150 MB per tab
Google Chrome	chrome.exe (Extension)	Background Extension	~80 MB each
WhatsApp	WhatsApp.exe (or WhatsApp)	Main App Interface	~200 MB

Task 4 :- Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

ANS :-

Steps :-

- 1. Download & Run : Install Advanced IP Scanner (Windows) or Angry IP Scanner (Mac/Linux/Windows).**
- 2. Scan : Click Scan to check your Wi-Fi network.**

Example Output :

IP Address	Device Name	Device Type
192.168.1.1	TP-Link_Router	Home Router
192.168.1.5	Android-Phone	Smartphone
192.168.1.9	Dell-Laptop	Personal Computer

Task 5 :- Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

ANS :-

Tool	When to Use It
Cable Tester	Use it to check if an Ethernet cable is broken, damaged, or wired incorrectly.
Crimping Tool	Use it to attach RJ45 connectors onto the ends of network cables.
Multimeter	Use it to check power levels, voltage, and electrical continuity in switches or power supply units.